

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 11: Multiplexing Techniques (FDM, TDM, WDM)

Course: Computer Networks (Credit Hours: 3)

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1. WHAT IS MULTIPLEXING? (SHARING ONE HIGHWAY)

Multiplexing is the technique of combining multiple separate data signals (from multiple senders) and transmitting them simultaneously across a **single physical medium/cable**.

Easy Everyday Example A highway with multiple lanes, or a train carrying hundreds of passengers in separate seats across one single railway track. Instead of building 10 separate physical cables, we combine 10 signals onto 1 high-speed line!

Key Hardware: **MUX (Multiplexer)** combines signals at sender side; **DEMUX (Demultiplexer)** separates them back at receiver side.

2. THREE PRIMARY TYPES OF MULTIPLEXING

1. FDM (Frequency Division)

Divides total channel frequency/bandwidth into smaller non-overlapping frequency bands.

- Used for **Analog signals**.
- Separated by small empty bands called **Guard Bands** to prevent overlap.
- **Example** Cable TV (all channels on 1 cable) & FM Radio.

2. WDM (Wavelength Division)

Variation of FDM designed specifically for **Fiber Optic cables**.

- Combines different signals as different **wavelengths (colors of light)**.
- Uses optical prisms to merge and split light beams.
- **Example** High-speed underwater fiber optic backbones.

3. TDM (Time Division)

Divides the transmission channel by **time slots** rather than frequencies.

- Used for **Digital data**.
- Each user gets 100% of the channel bandwidth, but only for a tiny slice of time.
- **Example** Cellular phone networks & digital telephone trunks.

3. SYNCHRONOUS TDM VS. STATISTICAL (ASYNCHRONOUS) TDM

Feature	Synchronous TDM	Statistical (Asynchronous) TDM
Time Slot Allocation	Pre-assigned fixed time slot to each user whether they have data or not.	Time slots allocated dynamically on-demand only to users who have active data to send.
Wasted Bandwidth	High (empty time slots are transmitted if a user is idle).	Zero waste (capacity is fully utilized by active nodes).
Addressing Overhead	No address needed (slot position determines receiver).	Address tag needed in each slot to identify source and destination.
Efficiency	Low efficiency during bursty data traffic.	Much higher throughput and efficiency.

4. SUMMARY COMPARISON: FDM VS. TDM VS. WDM

Technique	Domain Divided	Signal Type	Primary Medium
FDM	Frequency bands	Analog	Coaxial cables, Radio / Air
WDM	Light wavelengths (Colors)	Optical (Light pulses)	Fiber Optic Cables
TDM	Time slots	Digital	Twisted Pair, Digital trunks

5. QUICK STUDENT REVIEW QUESTIONS

Q1: What is the purpose of Guard Bands in FDM?

Ans: Guard bands are small unused frequency strips placed between adjacent frequency channels to prevent overlapping interference (crosstalk) between neighboring radio or TV stations.

Q2: Why is Statistical TDM more efficient than Synchronous TDM?

Ans: Because Synchronous TDM wastes time slots on idle users, while Statistical TDM only assigns time slots to stations that actually have active data packets waiting to send.